

Year 7 • Chapter 9 • The number plane: answers (teacher copy)

9.01 Grid references on maps (booklet p. 3)

WE DO examples

1. Ex 1: the lake
2. Ex 2: the library
3. Ex 3: D9
4. Ex 4: B2, C2, B3, C3
5. Ex 5: e.g. walk to Gauss Rd, west to Euclid St, then north
6. Ex 6: it is a convention so everyone reads it the same way

Set A

1. lake
2. library
3. stadium
4. student centre

Set B

1. D9
2. F2 or G2
3. A8 or B8
4. B2, B3, C2 or C3
5. H7, H8, I7 or I8
6. G10, H10 or I10

Set C

1. e.g. A6
2. e.g. E1
3. E6
4. main campus (row 10)
5. 4 squares
6. student centre and main campus

Set D

1. own map
2. own directions

Extension

1. C5
2. 18 squares

Exit ticket

1. Level 1: stadium
2. Level 2: F2
3. Level 3: student centre (H7)

9.02 Points in the first quadrant (booklet p. 7)

WE DO examples

1. Ex 1: plotted
2. Ex 2: plotted on the axes
3. Ex 3: P(4, 1), Q(2, 5)
4. Ex 4: No: (2, 5) is 2 across, 5 up
5. Ex 5: rectangle
6. Ex 6: (0, 0)

Set A

1. plotted
2. plotted

Set B

1. (1, 2)
2. (4, 5)
3. (7, 1)
4. (0, 3)
5. (5, 0)
6. (6, 4)

Set C

1. rectangle
2. isosceles triangle

Set D

1. (1, 6)
2. x-axis points have $y = 0$; y-axis points have $x = 0$

Extension

1. (3, 5)
2. 24 square units

Exit ticket

1. Level 1: 7
2. Level 2: on the y-axis, 5 up
3. Level 3: (1, 5)

9.03 The number plane (booklet p. 11)

WE DO examples

1. Ex 1: 1st top right, 2nd top left, 3rd bottom left, 4th bottom right
2. Ex 2: 3rd
3. Ex 3: A(2, 3), B(-3, 1), C(-2, -4), D(4, -2)
4. Ex 4: 4th
5. Ex 5: a zigzag line
6. Ex 6: e.g. (-2, 3); e.g. (0, 5)

Set A

1. 3rd
2. 1st
3. 2nd
4. 4th
5. y-axis
6. x-axis

Set B

1. (1, 3)
2. (-4, 2)
3. (-3, -3)
4. (3, -2)
5. (0, -4)
6. (-5, 0)

Set C

1. a flower / star shape (symmetrical)
2. D(-4, 3)

Set D

1. B: 2nd
2. (0, 0)
3. (-3, -2)
4. 7 units

Extension

1. 12 square units
2. (1, -1), (4, -1), (1, -3)

Exit ticket

1. Level 1: 2nd
2. Level 2: e.g. (3, -1)
3. Level 3: (-2, -5)

9.04 Plotting tables of values (booklet p. 15)

WE DO examples

1. Ex 1: vertical line $x = -2$
2. Ex 2: horizontal line $y = 3$
3. Ex 3: straight line going down
4. Ex 4: straight line going up steeply
5. Ex 5: y goes up by 2
6. Ex 6: $y = 7$

Set A

1. vertical line
2. horizontal line

Set B

1. (0, 5), (1, 4), (2, 3)
2. (-1, -1), (0, 1), (1, 3)
3. (1, 2), (2, 4), (3, 6)
4. (0, 1), (2, 1), (4, 1)

Set C

1. straight line going down
2. straight line going up

Set D

1. a U-shaped curve (parabola)
2. no, it is a curve

Extension

1. -5
2. $y = 4x$

Exit ticket

1. Level 1: (0, 3), (1, 5), (2, 7)
2. Level 2: horizontal line
3. Level 3: it goes up by 3

9.05 Tables of values from rules (booklet p. 19)

WE DO examples

1. Ex 1: 2, 3, 4, 5, 6
2. Ex 2: -11, -9, -7, -5, -3
3. Ex 3: 6, 5, 4, 3, 2
4. Ex 4: -5, -2, 1, 4, 7
5. Ex 5: $y = 2x + 4$
6. Ex 6: $y = 8 - x$

Set A

1. 2, 3, 4, 5, 6
2. -2, 0, 2, 4, 6
3. -4, -3, -2, -1, 0
4. -10, -5, 0, 5, 10

Set B

1. -11, -9, -7, -5, -3
2. 6, 5, 4, 3, 2

Set C

1. -5, -2, 1, 4, 7
2. 7, 5, 3, 1, -1
3. -4, -3, -2, -1, 0
4. 10, 7, 4, 1, -2

Set D

1. $y = 2x + 4$
2. $y = 8 - x$
3. $y = 4x - 3$
4. $y = -3x$

Extension

1. 2, 0, 0, 2, 6
2. $y = x^2$

Exit ticket

1. Level 1: 3
2. Level 2: -1, 3, 7
3. Level 3: $y = 3x + 1$

9.06 Graphing linear equations (booklet p. 23)

WE DO examples

1. Ex 1: $y: -3, -1, 1, 3$; straight line
2. Ex 2: line through (0, 3) going down
3. Ex 3: yes
4. Ex 4: yes
5. Ex 5: $y: -4, -3, -2, -1, 0$
6. Ex 6: (0, -1)

Set A

1. $y: -1, 0, 1, 2, 3$
2. $y: -3, -1, 1, 3$

Set B

1. no
2. no
3. yes
4. yes
5. yes
6. yes

Set C

1. $y: 5, 4, 3, 2, 1$
2. $y: -4, -3, -2, -1, 0$

Set D

1. parallel lines, 2 units apart
2. (1, 2)

Extension

1. $m = 3$
2. $y = 2x + 1$ and $y = 2x - 3$

Exit ticket

1. Level 1: 5
2. Level 2: yes
3. Level 3: yes